

Datasheet

13 April 2026

Motivation	The dataset was created by Imperial College for the “Capstone Black Box Optimisation” project, which is part of the following course: Professional Certificate in Machine Learning and Artificial Intelligence
Composition	There are eight datasets, each for a different function within the BBO project. The datasets range from 2 to 8 dimensions. The initial datasets contain 8 to 30 rows for the students to start with. The input values are all between 0 and 1. The output values are floating point numbers. Example data point from function 1 (a 2D dataset): Input row: 0.693878, 0.877551 Output: 1.3147995487661745e-46 There are no gaps in the dataset.
Collection process	The BBO project started on 29 January 2026, with one submission per week. Queries were generated by the students. My strategy was to use Google Gemini as an AI Modelling Assistant. Various models and acquisition functions were used, and these changed as the project progressed week by week.
Preprocessing	No pre-processing or transformations were applied.
Uses	This dataset is purely for educational purposes, and was created for the course. As this is a fictional dataset, there is no risk of inappropriate use.
Distribution	The initial dataset was provided by the course management to the students. The initial dataset is published on the GitHub repository for my project. https://github.com/BriarCloudview/Capstone1/tree/initial-data My final dataset will be similarly published at the end of the project.
Maintenance	The initial dataset is owned and maintained by the course management.

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